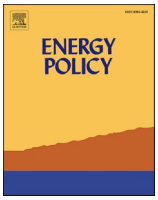
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Evidence of households' demand flexibility in response to variable hourly electricity prices – Results from a comprehensive field experiment in Norway

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Terminology:
Implicit demand flexibility: Consumers' voluntary response to*price signals*
Price response: Reduction of electricity consumption due to*price signals*
Price signal: Hourly electricity prices over one day specified by*peak price level*and*price profile*
Peak price level: Magnitude of the hourly electricity price in the*peak hours*of the*price signal*
Price profile: Shape of the*price signal*(e.g., number and duration of*peak hours*)
Peak hours: Hours with the highest electricity price of the*price signal*

ABSTRACT

Demand-side flexibility which reacts to hourly price signals is expected to play an essential role in balancing intermittent electricity generation in the future carbon-neutral power system. This study investigates the price responsiveness of households and various influencing factors through a pricing experiment which was implemented as a reward scheme and involved 3746 Norwegian households. These households, characterised by a highly electrified energy usage such as electric heating and charging electric cars, were subjected to variable hourly price signals with a one-day advance notification over the course of three winter months. The study reveals that households reduced their electricity demand by, on average, 2.92% in hours with high prices, and the reduction did not diminish significantly after repeated interventions. Moreover, an increased response could be observed for price signals with a short peak price period and when prices exceeded a threshold of 15 NOK/kWh. These results suggest that despite the limited potential of manual demand response from households, it can still be relied upon and utilised to enhance power system operations and planning. Additionally, to fully exploit its potential, it is recommended to enhance price responsiveness through tailored price information and by incentivising investments in automatic response and energy storage.

1. Introduction

Electricity consumers must play a more active role in the future energy system to balance variable renewable production and reduce congestions in the power grid (Yule-Bennett and Sunderland, 2022). An active consumer side allows a faster and economically efficient transition to a climate-neutral society. Furthermore, the importance of flexible consumers is emphasised by the need to reduce peak electricity demand in the ongoing European energy crisis due to a gas supply shortage (smartEn, 2022).

Flexible consumers may change their electricity consumption in response to variable electricity prices or grid tariffs. This kind of demand flexibility is also known as 'price responsiveness of demand' or 'implicit demand flexibility', commonly defined as consumers' voluntary response to price signals (European Smart Grids Task Force Expert Group 3, 2019; SEDC, 2016). Europe and Norway will experience greater electricity price variability towards 2035 due to a rapid increase of intermittent renewable energy production from solar and wind. In the even longer term, price variation will be dampened as the share of flexibility in the power system increases, but it will still be significantly higher than today (Statnett, 2023). Thus, this paper investigates the potential of implicit demand flexibility from Norwegian households through a large pricing experiment. This experimental study focuses on manual price responsiveness in winter months when exposed to hourly price signals with a notification one day ahead and the price signals were implemented with a reward scheme where households got compensated for their response.

In Norway, preconditions for implicit flexibility are already in place, which is beneficial for conducting this kind of study. First, electricity consumers must face variable electricity prices through either power contracts with variable electricity prices, grid tariffs, or both. In Norway, around 76% of households have power contracts tied to the hourly electricity market price at Nordpool,[1] and variable grid tariffs were introduced for all households in the summer of 2022. Second, it must be

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[1] At the time of the pricing experiment in early 2021, https://www.ssb.no/en/statbank/sq/10076907.

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possible to track electricity demand per household with at least the same time resolution, as the prices vary. Since 2019, smart meters with an hourly resolution have been in place for all Norwegian consumers. Third, households should have flexible electricity consumption to respond to variable prices. Most Norwegian households use electricity as their primary energy source for heating and hot water, and 16% of private cars are electric[2] and are, therefore, highly electrified. Space heating and charging of electric vehicles can potentially be controlled flexibly with little consequences for the users' comfort.

This paper targets households as they are at the centre of the EU's energy policy (European Commission, 2016) and are relatively homogenous compared to other consumer groups. Additionally, due to their high electrification, Norwegian households have a significant share of over 50% of the electricity demand in urban regions with little energy-intensive industry.[3]

Price responsiveness can be evaluated using historical data (Reiss and White, 2005) or experimental setup (Cappers and Spurlock, 2020; Frederiks et al., 2016). We decided to use an experimental setup since it allows for a control group, complete control over the price signals, and a pure focus on economic incentives independent from a specific use case or situation, which may affect the households' response. Furthermore, it allows for testing extreme electricity prices or economic incentives for reducing electricity consumption that seldom occur in real life and historically had not happened at the time of the experiment in early 2021.

This paper contributes to the research on residential price responsiveness by investigating particular issues such as factors with impact on the response, the nature of the response and possible side effects. For these issues, the existing body of literature often lacks a consensus. Additionally, this study demonstrates its novelty by incorporating three distinct features, the combination of which, to our knowledge, has not been used in existing studies. First, the experimental setup used a pure monetary incentive and did not assess a specific policy instrument such as dynamic electricity price contracts, grid tariffs or demand response programs. Thus, it avoided an instrument specific framing that may influence consumers' motivation to respond. Second, several price signals covering a broad range of price levels were tested within the same experimental setup which allows for a clear comparison of the response to these price signals. Third, the experiment was conducted with highly electrified households which are familiar with dynamic electricity price contracts and the results can therefore give valuable insights into the future potential of manual price response.

The structure of the paper is as follows. First, the state-of-the-art of implicit demand response is briefly reviewed to reveal gaps in the existing research body and to formulate research questions. Second, the experimental setup and the collected data are described. Third, the econometric models used to analyse these data are introduced. Results are presented in chapter 4 and their validity discussed in chapter 5. Finally, we conclude and formulate policy advice in the last chapter. The background chapter, the research questions, and the results chapter have the same order of the topics in their subsections to ease the paper's readability.

1.1. Background

An extensive research body on the demand response of households to price signals exists. Here, we focus on the magnitude of the response, factors that influence it, the nature of the response and possible side effects. The findings from this literature review are summarised by formulating the identified research gaps.

International and Norwegian studies have proved that households respond to various price signals with their electricity consumption. Reviews of mainly European pilots (Stromback et al., 2011; VaasaETT, 2017) reported a demand reduction in peak hours of between 9 and 24%, dependent on the type of price signal, and an average reduction of 9% without automation and 23% with automation. Based on international evidence, Faruqui et al. (2017) showed approximate average demand reductions of 10% and 20% without and with enabling technologies, respectively. Furthermore, dynamic pricing experiments in the US resulted in treatment effects of at least 7.5% (Harding and Sexton, 2017), whereas California's time-of-use pricing pilots gave a lower reduction of 1.4 to 3.6% in defined peak periods in the winter months (George and Bell, 2018). A recent meta-analysis (Buckley, 2020) argues that realistic reduction estimates are low with 1.9 – 3.9%. Norwegian studies have studied household demand response mainly from various grid tariff designs. A pilot study (Sæle, 2020) concluded that households made only minor changes to their electricity consumption when exposed to two different variable grid tariffs. Elvia, the largest distribution system operator, tested a time-of-use tariff with doubled tariff rates from 6 a.m. to 10 p.m. and found a 1.8% average demand reduction during these hours (Fors and Ahmed, 2021). Another distribution system operator tested critical peak pricing with nine peak days and substantially higher rates from 4 p.m. to 10 p.m. of 10 NOK/kWh versus 0.05 NOK/kWh in non-peak hours (Garnache et al., 2021). An average demand reduction of 13% in peak hours was achieved in this pilot.

The responses differed widely between studies, and differences can partly be explained by the type of price signal, especially the peak price level, and other factors such as duration and the frequency of peak price periods. Knowledge about how these factors affect the response is very relevant since these design parameters can be decided on in grid tariffs or demand-response programmes. Several reviews (Chitkara et al., 2016; Faruqui et al., 2017; Stromback et al., 2011) concluded with a clear positive relationship between the achieved demand reduction and the peak price level measured as the peak to off-peak price ratio. A price threshold may seem to exist at a price ratio over 12 times or rebates over 0.8 €/kWh where the response increases additionally (Stromback et al., 2011). However, a weakness of this analysis is that peak period duration and frequency were not included as explanatory factors. These factors are usually correlated to peak price levels since peak period prices are typically higher for price signals on fewer and shorter occasions. This may also explain why other studies that tested different hourly prices (CER, 2011; Gattaciecca et al., 2020; Gillan, 2017; Ito et al., 2018) found a marginal effect on the response or even none of higher prices (Guo et al., 2019). The same criticism can be applied to results stating that a long peak duration or high frequency may lead to less response (Chitkara et al., 2016; Stromback et al., 2011). Furthermore, some studies showed that a low frequency of peak events could lead to a slightly higher response but that events on two consecutive days did not affect the response (Gattaciecca et al., 2020; Schofield et al., 2015).

The response to price signals also depends on external factors such as the outdoor temperature, especially if electricity is used for heating. The thermal storage capacity of the building body, or alternative heating sources such as wood stoves, allows for electricity demand flexibility and should, in consequence, lead to a higher response potential on colder days. This expectation is supported by results from various studies that showed a higher response on cold days than on warmer ones (Hofmann and Lindberg, 2019; Hu et al., 2015; Martin and Rivers, 2018; Schofield et al., 2015). However, a Norwegian study (Hofmann and Lindberg, 2019) also showed that the price elasticity was reduced for very cold temperatures below -15 °C. A proposed explanation was the lack of substitutes for electrical heating since all available energy sources, including the substitutes, are needed to keep the indoor temperature at the desired level.

A persistent response in comparable situations, for example, with the same prices and outdoor temperature, is a precondition for using implicit demand flexibility in peak reduction programmes or for including

[2] End of 2021, https://www.ssb.no/en/transport-og-reiseliv/landtransport/statistikk/bilparken.

[3] Calculated based on data from Elhub for the first quarter of 2021, https://elhub.no/statistikk/stromforbruk/.

it in demand prognoses models. Most empiric evidence (CER, 2011; Gillan, 2017; Ito et al., 2018; Parrish et al., 2019; Stromback et al., 2011) has concluded that the response is relatively stable over time up to 24 months for pricing pilots and up to 6 years for a electricity distribution tariff (Bartusch and Alvehag, 2014). However, some studies saw a slight reduction in response (Gattaciecca et al., 2020), while others concluded with an increase or both effects (Fowlie et al., 2020; Parrish et al., 2019; Stromback et al., 2011). A declining response over time is usually explained by response fatigue where persons get tired of following the price signals and to adjust their electricity consumption accordingly (Kim and Shcherbakova, 2011). This argument is clearly supported by a dynamic pricing pilot where strong response fatigue was observed for manual response and none for automatic response (Vanthournout et al., 2015).

Not only should the response be persistent over time, but it should also be stable at the hourly level in the peak price period to get a predictable response. The results were not conclusive for this question. While some research suggested a limited effect of the time of the day (Gattaciecca et al., 2020), others found a clear correlation between response and consumption level, which corresponds to the time of the day electricity is mainly used, and people are at home (Schofield et al., 2015; Zhou et al., 2017).

If households respond to price signals with load shifting, rebound effects can occur, possibly leading to higher load peaks. It is, therefore, relevant to know if households either reduce or shift their demand and if it leads to a rebound in electricity consumption before or after peak price hours. It seems that the majority of pricing pilot results have indicated that no load shifting and rebound occurs (Di Cosmo and O'Hora, 2017; Gattaciecca et al., 2020; Harding and Sexton, 2017; Lanot and Vesterberg, 2021) or that the electricity consumption is even reduced in hours before and after the treatment hours or the following days (Garnache et al., 2021; Ito et al., 2018; VaasaETT, 2017). However, some research also had contrary results and saw load shifting and accompanying rebound effects in their studies (Azarova et al., 2020; CER, 2011; Schofield et al., 2015).

In summary, the literature review indicates that households do respond to price signals, although the extent of their response varies significantly. Price signal characteristics, such as the peak price level, the duration of peak prices, and the frequency of events, may account for some of this variability. However, these factors are often studied in isolation rather than collectively, making it challenging to attribute specific impacts to each of these elements. In addition, the literature is inconclusive regarding the persistence of the response when exposed to price signals over a longer period, the rebound effect, and the hourly stability of the response in peak hours. Our study contributes to filling these identified research gaps in residential price responsiveness, particularly in areas where there are inconclusive results regarding the influence of various factors on the response. Furthermore, it investigates the potential of manual implicit demand response in a Norwegian context for several climate regions.

1.2. Research questions

The paper aims to estimate the relation between hourly variable electricity prices and electricity consumption, i.e., manual implicit demand response from highly electrified households in cold climates. The research questions which are answered by this study are grouped into three categories:

- Q1: How large is the aggregate demand response from households to varying hourly electricity prices?
  a) What is the primary type of response, load reduction, or load shifting?
  b) How much is electricity consumption reduced in hours with exceptionally high prices and how does the response vary for the different price signals?
- Q2: What factors affect the price response?
  a) What is the impact of the price signal's characteristics on the response?
  b) How do external factors such as outdoor temperature affect the response?
  c) Is the response persistent over time, or does it increase due to learning or decrease due to response fatigue? Can long-lasting changes in electricity consumption habits be observed?
- Q3: What is the nature of the observed response and its side effects?
  a) How stable is the hourly response in periods with high prices?
  b) Does a rebound effect due to load shifting occur?

2. Experimental design

Our experiment was conducted during three winter months in 2020/21 and is part of the Norwegian transmission system operator Statnett's research project iFlex. The iFlex project consists of two phases: the winter season in 2019/20 and 2020/21. Our analysis focuses solely on the second phase with a full-scale experiment, whereas results from the first phase can be found in Hofmann and Lindberg (2021). The main facts of the experiment and data collection are summarised in Table 1.

The pricing experiment was implemented as a randomised controlled trial, which is the preferred method for obtaining estimates for demand reduction due to price signals (Cappers, 2013; Cappers and Spurlock, 2020; Frederiks et al., 2016) and is widely used in recent studies (Azarova et al., 2020; Burkhardt et al., 2019; Gattaciecca et al., 2020; Gillan, 2017; Ito et al., 2018; Martin and Rivers, 2018). In a randomised controlled trial, the participants are randomly assigned to either treatment groups exposed to price signals or to a control group. Consequently, all differences in electricity consumption between the groups should be due to the price signals. The economic incentive to respond to the price signals was implemented with a reward scheme wherein households in the treatment groups received rewards when they reduced their electricity consumption. A detailed description of data collection process and the complete iFlex dataset are provided in an accompanying data article (Hofmann and Siebenbrunner, 2023).

2.1. Recruiting and randomised group assignment

Participating households were recruited in November 2020 from the customer base of three electricity suppliers covering the major Norwegian city regions with various climates: Oslo, Bergen, Trondheim, Bodø, and Tromsø. Each electricity supplier sent an email invitation to all of its customers within their respective regions. To increase participation, households could win a smartphone or free electricity for one year, depending on the recruiting electricity supplier. Opt-in recruitment always suffers from the threat of self-selection bias. Therefore, a particular

Table 1
Summary of the experiment's characteristics.

Participating households:	3746
Pricing experiment	
Time period:	16 December 2020 to 25 March 2021
Price signals tested:	14
Number of treatment groups:	11
Sum of experiment days of all treatment groups:	254
Electricity data	
Time period:	1 December 2020 to 26 March 2021
Hours excl. weekend and holidays:	1776
Average consumption:	2.52 kWh/h
Survey data	
Time period:	12 May to 10 June 2020
Number of questions:	43
Responses from project participants:	1733
Responses from the population:	1800

emphasis was put on having a recruitment process that was as easy as possible, and a neutral invitation text to avoid attracting only participants interested in demand response. The recruitment resulted in 4339 of around 45,000 invited households and a final number of 3746 households which our analysis is based on.

The random group assignment of the households was conducted per city, aiming at a group size of at least 200 households. It resulted in 11 treatment groups and five control groups, one for each region to control for local conditions. Due to the lack of participants in Bergen, the control group includes 59 households recruited in the first season. A comparison of the average hourly electricity consumption of each region's control and treatment groups before the experiment started shows differences of up to 10%. However, we may assume a successful random assignment since the differences are not statistically significant. Both the group sizes and the electricity consumption comparisons can be found in appendix Table 8.

2.2. Economic incentive and price signals

The various price signals gave an economic incentive to the households to reduce their electricity consumption on experiment days. However, the experiment's price signals did not replace the households' power contracts which in Norway are usually real-time pricing contracts tied to the hourly electricity spot price. Instead, if a household responded to a price signal, they could earn a reward that reflected the household's savings on the electricity bill if these prices had occurred in real life. As a result, the economic incentive to reduce electricity consumption in peak price hours should be equivalent to a price. Nevertheless, due to behavioural phenomena of loss aversion (Kahneman et al., 1991), a reward may not be perceived in the same manner as savings and possible implications for the validity of the results are discussed in chapter 5.

This experimental setup resembles critical rebate programs (Energy Technologies Area and Berkeley Lab, n.d.; Faruqui and Bourbonnais, 2020; VaasaETT, 2017) and was selected for two specific reasons. Firstly, it ensures a sufficiently high recruitment rate of households into the experiment. Convincing households to participate would have been challenging if they had to switch their existing power contract to an experimental one with exceptionally high prices on experiment days. Furthermore, the households who would have opted in on these terms would introduce a significant self-selection bias and threaten the external validity of the results. Secondly, the rebate setup provides a high degree of flexibility in designing the price signals, allowing for testing of extraordinarily high prices.

The rewards were paid at the project's end as a rebate on the electricity bill. The rebate was computed as the sum of the hourly experiment prices and the change in electricity demand during the respective hours. Households could have a negative reward value, but it was adjusted to zero at the end to ensure that households did not receive an increased electricity bill. A necessity for calculating the change in electricity consumption is the estimation of the counterfactual electricity consumption without a price signal of each individual household. This baseline was calculated by using the average electricity consumption of the last 10 non-experiment days for the same hour of the day, adjusted for systematic differences between weekdays and changes in outdoor temperature. Similar baseline methods are implemented in commercial products such as PGE's Peak Time Rebates program in Oregon.[4] Moreover, a test using historical electricity consumption data from the participating households demonstrated that the method generated individual household baselines with only minor deviations. The primary reason for selecting this approach was its simplicity, which also made it possible for participants to understand. It is essential to clarify that the baseline approach was exclusively utilised for the computation of reward values per household and was not used to assess the observed price response.

The treatment groups were exposed to various price signals on up to 21 experiment days from 16 December 2020 to 25 March 2021. Price signals were designed before the experiment started, while the selection of experiment days was determined dynamically one week before based on weather forecasts. The objective was to include the coldest days, which typically have high electricity consumption due to electric heating. These specific days were chosen because knowledge about achievable demand reductions is of greater interest when electricity demand is high. Each treatment group received one of four predefined price signals on each experiment day. The respective price signal was mainly randomly selected, but it was assured that each price signal was tested equally often in each treatment group, around five times.

The *price signals* consist of hourly electricity prices that vary over one day. They are designed using two parameters, the *peak price level* in the peak hours and the specification of the peak hours by the daily *price profile*, as visualised in Fig. 1. The peak price level varies from 2 NOK/kWh to 30 NOK/kWh, and prices outside the peak periods are set to 1 NOK/kWh in all profiles besides P0. Profile P0 is equal to profile P but has zero prices in non-peak hours, which allows us to investigate if non-peak hours without the possibility of getting a reward for electricity reductions affect the response in the peak hours. To put the prices into perspective, the total electricity price for households, including grid tariffs and taxes, was historically around 1 NOK/kWh[5], and the maximum allowed clearing price in the Nordpool day-ahead market is 52 NOK/kWh.[6] All price profiles were designed so that the peak hours align with the historical demand pattern of an average household and the typical fluctuations in hourly spot prices over a day. This approach ensures that peak hours correspond to periods of high electricity demand and higher spot prices. The rationale behind is that price signals in real life originate from grid tariffs or from real-time electricity price contracts reflecting the day-ahead market which typically have higher prices when demand is high. The duration of the peak hours varies between profiles to be able to test the effect on the demand response. Profile B has the longest peak period from 7 a.m. to 8 p.m., while profiles A and C have two separate peak periods in the morning and afternoon. Profiles P and P0 have a short peak period of 2 h between 4 and 6 p.m.

Treatment groups were informed about the occurrence of an experiment day via an SMS or a push message in an app at 6 p.m. the day ahead. Detailed information about the hourly prices and advice on how to reduce electricity consumption in peak price hours were available on either a web page or in the app.

2.3. Data

All data for this study were collected in the winter season of 2020/21. The dataset consists of hourly electricity consumption data, surveys answers, web page visits and outdoor temperature.

Hourly electricity consumption data are available from 3746 households from five Norwegian regions and were collected some weeks upfront and during the pricing experiment period covering 1 December 2020 to 26 March 2021, whereas the experiment period started 16 December 2020. The experiment regions were chosen to ensure that the whole country was covered, and thus results are valid for all households in Norway. Weekends and holidays were excluded from the data since the pricing experiment was only conducted on working days.

A survey was sent to the experiment households and households sampled from the general population in the respective regions after the

[4] https://portlandgeneral.com/save-money/save-money-home/peak-time-rebates.

[5] https://www.ssb.no/en/statbank/sq/10078603.

[6] https://www.nordpoolgroup.com/en/trading/Operational-Message-List/2022/04/day-ahead-new-harmonised-maximum-clearing-price-from-delivery-day-wednesday-11th-may-20220411122700/.

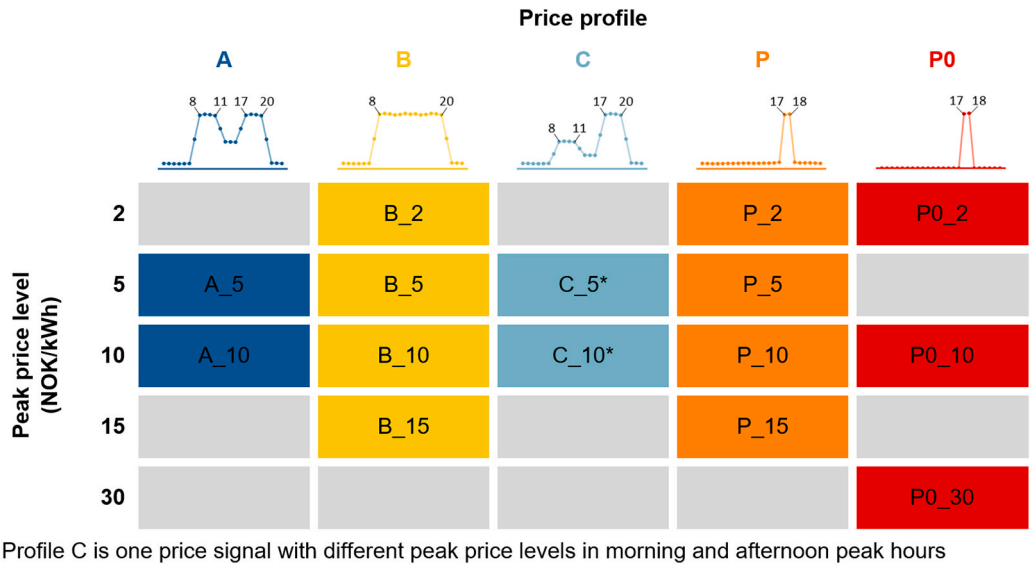


Fig. 1. Overview of price signals as a combination of peak price levels and 24-h price profiles. The start and end of peak periods are indicated with the hour of the day in the profile visualisations. Coloured boxes mark price signals that were used in the pricing experiment.

pricing experiment period. The questions covered socio-demographic variables, housing characteristics, electricity-consuming appliances, willingness to adjust electricity consumption, and the measures to respond to experiment price signals for the treatment groups. Around 47% of the experiment households answered the survey.

Table 2 summarises some selected characteristics of the households and shows that the experiment sample suffers from a self-selection bias, with fewer households living in apartments than in the population sample. This also explains the larger living space area, electric vehicle ownership, heat pumps, and electric warm water boilers. Even though the differences are statistically significant, the impact on the representativity of experiment results and external validity should be minor since the differences are small.

Additionally, the number of visits to the web pages containing the experiment prices were gathered. Two of the three electricity suppliers provided the data only an aggregated level. As Table 3 shows, the visits declined over the course of the pricing experiment.

Furthermore, hourly outdoor temperature data for each of the five regions were acquired from the Norwegian Meteorological Institute.[7] Each region is represented by one weather station, and the details are available from the data article (Hofmann and Siebenbrunner, 2023).

Table 2
Comparison of selected characteristics of households participating in the experiment and households sampled from the general population.

	Experiment households	Population	Difference	p-value
Answers	1733	1800		
Apartment	34%	40%	-6%	0.001**
Living space in m²	119	110	9	0.000***
Electric vehicle	18%	14%	4%	0.001**
Direct electric heating	55%	57%	-2%	0.386
Heat pump	32%	26%	6%	0.000***
Electric warm water boiler	76%	64%	12%	0.000***
People per household	2.4	2.3	0.1	0.024*
Household with children	25%	20%	5%	0.000***
Low income	7%	9%	-2%	0.091
No willingness/ ability to reduce consumption	3%	7%	-4%	0.000***

***p < 0.001, **p < 0.01, *p < 0.05.
Note: Answers were weighted accordingly to the distribution of the experiment households over the different regions. The p-value of the difference was calculated with Welch's *t*-test and two-proportions z-test.

Table 3
Visits to the web pages with information about the prices during the experiment period for two regions.

Region	Web page visits	Jan 2021	Feb 2021	Mar 2021
Oslo, Bergen, Trondheim	per month	6349	4101	2099
Tromsø	per month	1129	340	361
	maximum per day	170	70	59
Bodø	no data available			

3. Econometric analysis model

Econometric models are employed to analyse the response to the price signals. First, we examine whether the response is visually detectable in the electricity consumption data. Subsequently, we provide a detailed description of the econometric model specifications and the method for calculating percentage response values.

The visual assessment of electricity consumption on experiment days in Fig. 2 a) does not show a clear reduction in electricity usage during peak hours, when comparing treatment and control group. However, when accounting for underlying systematic differences, here the average electricity consumption between the groups, a response can be detected, as shown in Fig. 2 b).

The households' demand response is analysed with a fixed effect model for panel data to account for systematic differences in electricity demand. Fixed effect models have been widely used for estimating the treatment effect in recent pricing experiments (Azarova et al., 2020; Burkhardt et al., 2019; Garnache et al., 2021; Ito et al., 2018; Martin and Rivers, 2018; Wang et al., 2020). The model specification for explaining the hourly electricity consumption E_{it} of each household i in hour t in our analysis is presented in equation (1).

[7] https://seklima.met.no/.

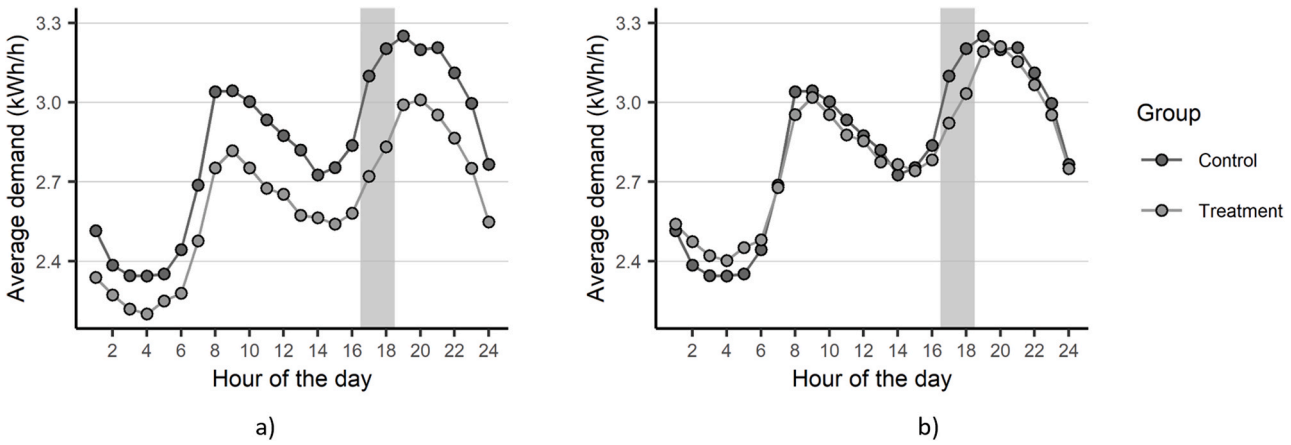


Fig. 2. Control and treatment group electricity demand in Oslo on experiment days for price signal P0_30 with a 30 NOK/kWh price level in peak hours 17 and 18 (highlighted in grey) visualised with a) observed data and b) data adjusted for differences in mean electricity consumption.

$$E_{it} = \beta_0 + \beta_1 D_{it}^{notreat} + \beta_2 D_{it}^{nopeak} + \beta_3 D_{it}^{peak} + \gamma_i + \delta_t + \varepsilon_{it} \quad (1)$$

Individual fixed effects for each household γ_i and time fixed effects for each timestep δ_t are included in the model formulation. Usually, household fixed effects control only for differences in the average electricity consumption between households (Azarova et al., 2020). However, since households can have different load profiles, for example, district heating versus electrical space heating, household fixed effects are specified as the hour of the day for each household, resulting in 24 fixed effects per household, see also (Martin and Rivers, 2018). Outdoor temperatures, daily routines and other external influences affect electricity demand and differ between the experiment regions. Hence, time fixed effects are specified for each hour of the experiment period and for each region, resulting in 1776 time fixed effects per region. In addition, the standard errors are clustered on household levels to allow for serially correlated demand observations per household.

Dummy variables indicate treated households and four types of hours, e.g., peak hours D_{it}^{peak}, non-peak hours on experiment days D_{it}^{nopeak}, hours on non-experiment days $D_{it}^{notreat}$ after the first experiment day, and hours before the start of the experiment. The change in electricity consumption is captured by the respective β for each of these dummies, with a main interest in the peak hours. Furthermore, the basic model specification (1) is adjusted to investigate the effect of various factors on the response. The model specifications are presented in Table 4.

In the model specification (2), the peak hour dummy variable D_{it}^{peak} is interacted with variables of interest D_{it}^{x} to understand their effect on the response in peak hours. Model (3) allows for estimating the difference in response by dropping one dummy variable of D_{it}^{x} which represents the reference. In addition, model (3) can incorporate confounding variables, which would cause multicollinearity issues in model (2). Model (4) has only one dummy variable for all hours on an experiment day D_{it}^{exp} instead of one for peak and one for non-peak hours. All the variables are described in Table 5.

The estimates obtained by all models are in kilowatt-hours per hour. However, calculating the percentage change in electricity consumption is often more informative for comparing results with other studies and for intuitively understanding the results without prior knowledge of the average electricity consumption. Many studies have used a log transformation of the electricity demand to obtain percentage coefficient estimates (Garnache et al., 2021; Ito et al., 2018; Wang et al., 2020). However, these estimates represent an unweighted average of the percentage change of each household, whereas we are interested in the aggregated change. Therefore, we divide the coefficient estimate β by the average electricity consumption across all households E_{β} in these hours to obtain percentage changes, as formulated in equation (5).

Table 4
Various model specifications for the analysis of the price response.

Model specifications for detailed response in peak hours		
Absolute	$E_{it} = \beta_0 + \beta_1 D_{it}^{notreat} + \beta_2 D_{it}^{nopeak} + \beta D_{it}^{peak} D_{it}^{x} + \gamma_i + \delta_t + \varepsilon_{it}$	(2)
Difference	$E_{it} = \beta_0 + \beta_1 D_{it}^{notreat} + \beta_2 D_{it}^{nopeak} + \beta_3 D_{it}^{peak} + \beta D_{it}^{peak} D_{it}^{x} + \gamma_i + \delta_t + \varepsilon_{it}$	(3)
Model specification for detailed response on experiment day		
Absolute	$E_{it} = \beta_0 + \beta_1 D_{it}^{notreat} + \beta_2 D_{it}^{exp} D_{it}^{x} + \gamma_i + \delta_t + \varepsilon_{it}$	(4)

Table 5
Variable definitions for estimating different treatment effects.

Variable	Short name	Description	Used in model
D_{it}^{exp}	Experiment hours	All hours on experiment days for treated households	(4)
D_{it}^{peak}	Peak hours	Peak hours on experiment days of treated households	(1), …, (3)
D_{it}^{nopeak}	Non-peak hours	Non-peak hours on experiment days of treated households	(1), …, (3)
$D_{it}^{notreat}$	Non-treatment hours	Hours on non-experiment days after the first experiment day of treated households	(1), …, (4)
D_{it}^{x}		Variable indicates that a household is in a treatment group and a time-specific characteristic	(2), …, (4)
$D_{it}^{psignal}$	Price signal	Dummies covering all price signals as a combination of peak price level and profile. Profile C was split into two dummies for morning peak period C_5 and afternoon C_10.	(2)
D_{it}^{price}	Peak price level	Dummies for each peak price level (2, 5, 10, 15 and 30 NOK/kWh)	(3)
$D_{it}^{profile}$	Price profile	Dummies for each price profile (A, B, C, P and P0)	(3)
D_{it}^{temp}	Temperature	Dummies with bins of the average outdoor temperature of the last 24 h with breaks at -20, -15, -10, -5, 0, 5 and 10 °C	(2), (3)
D_{it}^{expday}	Experiment day	Dummies with the consecutive number of the experiment day in each treatment group (1., 2.-5., 6.-10., 11.-15., 16.-21.)	(2)
D_{it}^{days}	Days without experiment	Dummies with the number of days without experiment before the actual experiment day (0, 1–6, 7–19)	(2)
D_{it}^{hour}	Hour of the day	Dummies for each hour of the day (1, …, 24)	(4)

$$Percentage\ change\ (\%) = \frac{\beta}{E_\beta - \beta} \quad (5)$$

4. Price response results

The results are organised in the same order as the research questions they address. First, the type of response and the observed reductions in peak hours are examined. Second, factors affecting price response are analysed such as price signal characteristics, outdoor temperature, and the response persistence over time. Third, the hourly response stability and the presence of rebound effects are investigated. The results are expressed as percentage changes in electricity consumption as described in the preceding chapter, and the absolute coefficient estimates from the econometric models are provided in the appendix.

4.1. Self-reported type of response: load reduction or shifting

After the experiment, the households were asked in the survey how they responded to the price signals. Fig. 3 shows the self-reported measures taken by households to reduce electricity consumption in peak hours, categorised into two types: load shifting and load reduction. These answers are from the subset of households, accounting for 48%, who confirmed that they actively responded to the price signals. Typical measures encompass shifting activities to other hours, such as laundry or dishwashing, as well as reducing electrical space heating by lowering indoor temperatures or using wood stoves as an alternative heat source. Only 16% of households answered that they shifted the electric vehicle charging. However, when considering households with electric vehicles, this measure was the most prominent with 81% using it in this subgroup.

4.2. Demand reduction in peak hours

The average electricity demand in peak hours across all price signals is statistically significantly lower, with a reduction of 0.085 kWh/h or 2.92% compared to the control groups. This result is based on model specification (1) as presented in Table 6. In addition, around half of the households self-reported in the conducted survey that they responded to the price signals. The average reduction result of a few percentages is low compared to international review studies (Faruqui et al., 2017; Stromback et al., 2011; VaasaETT, 2017; Zhu et al., 2018) and a recent Norwegian peak pricing pilot (Garnache et al., 2021) which reported demand reductions between 9 and 13%. However, other pricing pilots

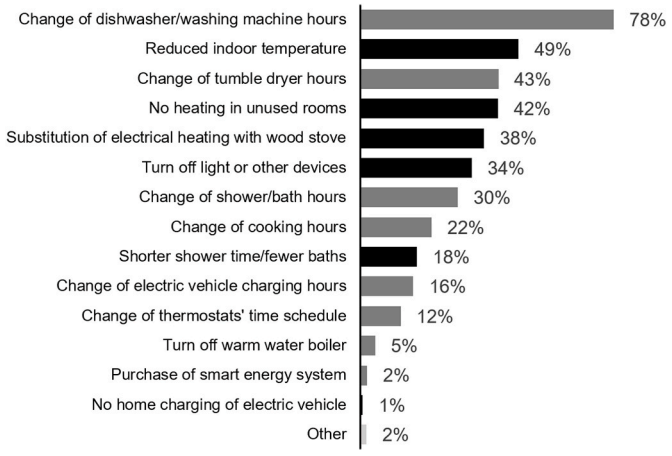


Fig. 3. Self-reported measures that the households undertook to reduce (black) or shift away (dark grey) electricity consumption in peak price hours as answered to the survey (N = 817).

Table 6
Absolute (kWh/h) and percentage change (%) in electricity consumption for peak, non-peak, and non-treatment hours across all price signals.

Variable	Change (kWh/h)		Change (%)		Mean consumption E_β (kWh/h)
	Estimate β	CI 95%	Estimate	CI 95%	
Peak hours	-0.085***	[-0.117, -0.052]	-2.92***	[-3.98, -1.81]	2.821
Non-peak hours	-0.058***	[-0.089, -0.027]	-2.16***	[-3.27, -1.02]	2.631
Non-treatment hours	-0.039**	[-0.063, -0.014]	-1.57**	[-2.52, -0.57]	2.440
R^2 adjusted	0.795				
N	6,652,896				

***p < 0.001, **p < 0.01, *p < 0.05.
Note: Estimates are calculated with model (1) and represent the average across all hours and price signals.

conducted in California (George and Bell, 2018), a Norwegian tariff pilot (Fors and Ahmed, 2021) and a meta-analysis (Buckley, 2020) showed reductions of similar magnitude.

In addition to the average response in peak hours, it is interesting to distinguish the response for each price signal $D_{it}^{psignal}$. Fig. 4 shows, based on model (2), that the households reduced their hourly electricity demand between 2.4% and 6.7% in peak hours, depending on the price signal. It can be observed that the peak price level in profiles A, B, and C has only a minor impact on the achieved demand reduction, whereas it seems to have an impact in profiles P and P0.

Price profile P0 has zero prices in non-peak hours, meaning that households received rewards solely for reducing their consumption during the two peak hours on experiment days. In contrast, profile P rewarded reductions in both peak and non-peak hours. A comparison of the price response in peak hours between these two profiles indicates no significant differences, as shown in Figs. 4 and 5 a). It seems that the prices in non-peak hours have no influence on the price response in peak hours.

4.3. Impact of price signal characteristics: Price profile, peak price level and event frequency

The characteristics of the price signals may influence the response. In the following, we investigate how the price profile, and thus implicitly the duration of the peak periods, the peak price level and the frequency of price signal events affect the response.

Fig. 5 a) shows that a shorter peak period results in a higher response. In detail, profiles P and P0, with peak duration of 2 h, increase the response by around -1.5 percentage points compared to profiles A, B, and C, with peak durations between eight and 13 h. No statistically significant difference in response can be observed if peak prices last for one period, as in profile B, or are split into a morning and afternoon period, as in profiles A and C. Previous research has reported the same effect of the peak duration with a higher response for shorter periods (Chitkara et al., 2016; Stromback et al., 2011).

Regarding the peak price level, Fig. 5 b) shows that a statistically significant increased response of -2.3 percentage points can first be observed for 15 NOK/kWh and above. Below this value, it seems that households react foremost to the existence of a price increase and not its magnitude. This finding is consistent with prior research that has reported only a marginal effect of price levels on the response (CER, 2011; Gattaciecca et al., 2020; Gillan, 2017; Guo et al., 2019; Ito et al., 2018), but challenges the clear relationship between price and response seen in other studies (Chitkara et al., 2016; Faruqui et al., 2017; Stromback et al., 2011). Furthermore, a study has also observed a threshold values

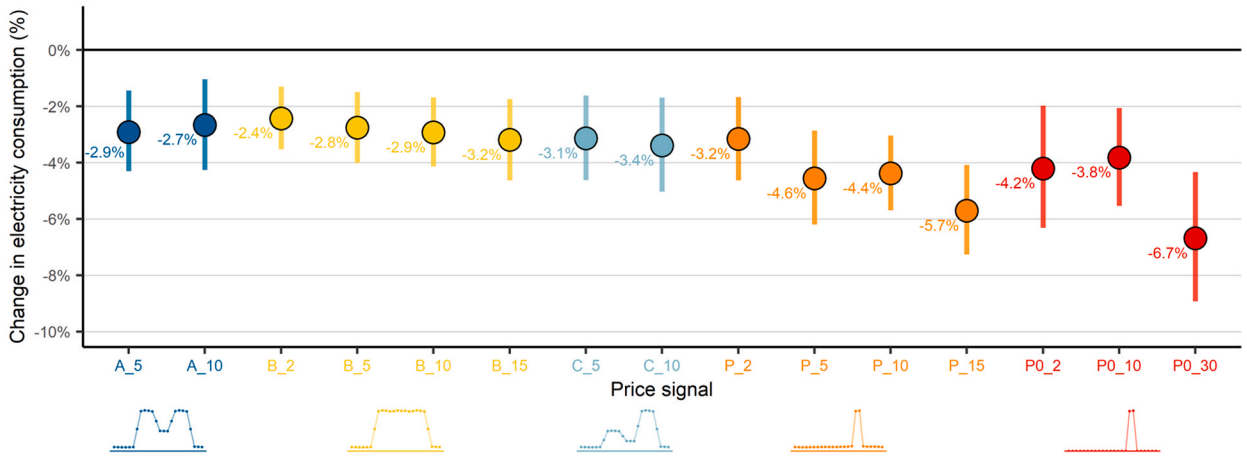


Fig. 4. Average hourly price response in peak hours for each price signal. Uncertainty of the estimates (point) is illustrated with the 95% confidence interval (line). Model results can be found in appendix Table 9.

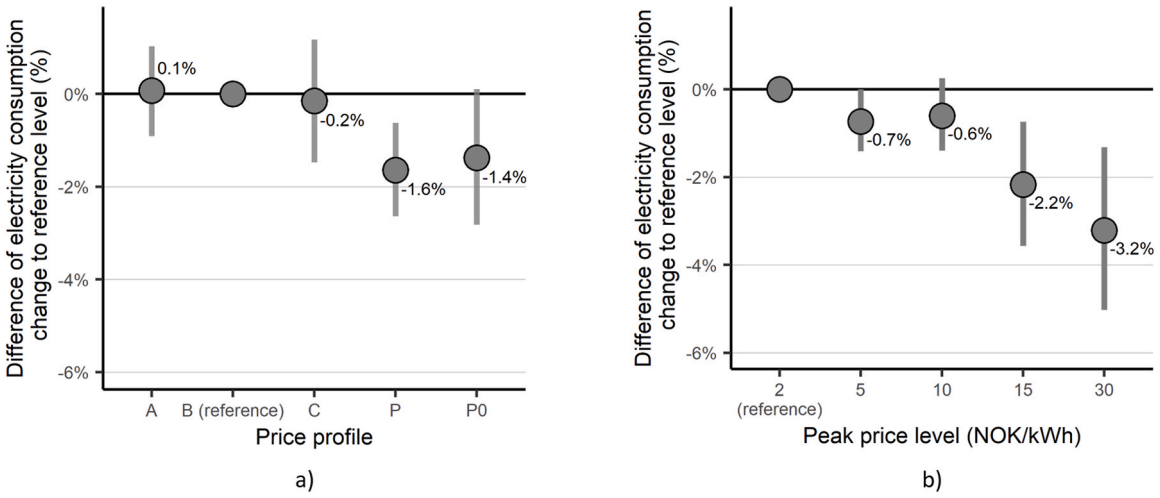


Fig. 5. Average hourly price response in peak hours with a 95% confidence interval for various price profiles and peak price levels compared to the reference of profile B2 and 2 NOK/kWh, respectively. The results are controlled for peak price level and price profile and can be found in appendix Table 11.

for increased response with values between 8 and 12 NOK/kWh<u>[8]</u> (Stromback et al., 2011). A possible explanation for our results may be that low-cost demand reduction measures are used first and that further response comes at a higher cost, for example due to discomfort.

Model specification (3) is used with the price profile $D_{it}^{profile}$ and peak price level D_{it}^{price} as confounding variables to obtain these results. Both variables have to be included in one model to consider that not all combinations of price level and profile are included in the experiment, as shown in Fig. 1. The results of this model give the difference in the response to a reference which is profile B and 2 NOK/kWh for price profile and price level, respectively.

A higher event frequency, meaning shorter periods without experiment days D_{it}^{days}, appears to result in a slightly lower response, as shown in Fig. 6. However, the differences in response are not statistically significant. The response results are based on model (2) and estimated for three cases: a consecutive experiment day, less than a week, and more than a week since the last experiment day. Our findings are in line with another study that reported a slightly higher response when peak events

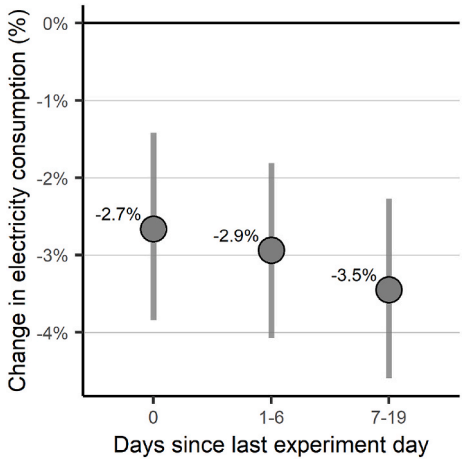


Fig. 6. Average hourly price response in peak hours with a 95% confidence interval in dependency of days without experiment before the actual experiment day. The first experiment day is excluded. Model results can be found in appendix Table 14.

[8] Calculated with an exchange rate of 1 Euro = 10 NOK. They reported 0.8 €/kWh for rebate programs and a price ratio of 12 for pricing pilots.

occurred once per week compared to several times per week (Gattaciecca et al., 2020). An explanation for these results could be response fatigue if peak events occur too frequently.

4.4. Impact of external factors: Outdoor temperature

Earlier studies of the price elasticity of Norwegian electricity demand revealed that the response was dependent on outdoor temperature (Hofmann and Lindberg, 2019). Model (2) together with intervals of the average outdoor temperature of the last 24 h D_{it}^{temp} was used to analyse this effect. A moving average of the temperature was selected to account for the thermal inertia of the buildings (Lindberg et al., 2019; Stokke et al., 2010). As shown in Fig. 7, the estimated response in peak hours for temperatures from -20 to 5 °C is stable. However, the price response is not statistically significant for temperatures below -10 °C, which may result from the limited number of observations for these cold temperatures. In addition, no response can be observed for temperatures between 5 and 10 °C. This influence of outdoor temperature on price responsiveness with a stronger response on cold days is supported by findings in other studies (Hofmann and Lindberg, 2019; Hu et al., 2015; Martin and Rivers, 2018; Schofield et al., 2015).

4.5. Impact of human factors: Persistence over time and changed consumption habits

It is essential to understand whether the price response is persistent over time or at least stabilises at a level. Two effects could affect the response. First, households could experience response fatigue over time since all electricity consumption adjustments are performed manually, leading to a lower response. Second, they could learn how to perform response measures with little effort, leading to the opposite effect. We have no data that would allow us to distinguish these two effects, but we will analyse how the response develops over the experiment duration of three months and how often the treatment households have checked the web pages with experiment price information.

It seems that, as presented in Fig. 8, the response is largest on the second to fifth experiment day with a 3.4% reduction and stabilises at 2.6% for the eleventh experiment day and onwards. The differences between the lowest and highest response are also statistically significant. The response estimates were obtained with model (2) and a dummy variable representing the consecutive number of the experiment day D_{it}^{expday} which measures time since experiment start.

The number of web page visits fell approximately 65% towards the last experiment month compared to the first month, based on the data in

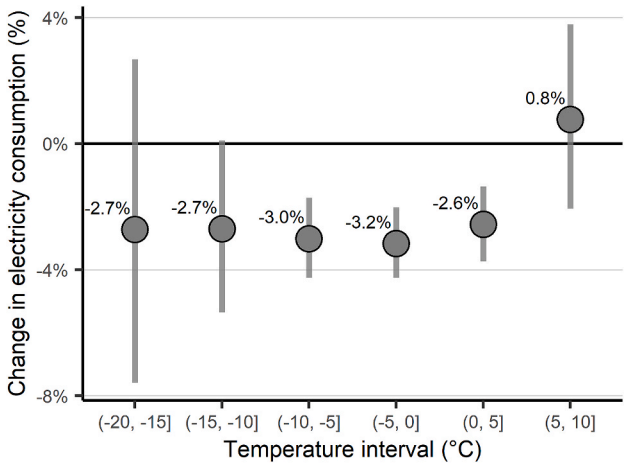


Fig. 7. Temperature-dependent average hourly price response in peak hours with a 95% confidence interval. Model results can be found in appendix Table 12.

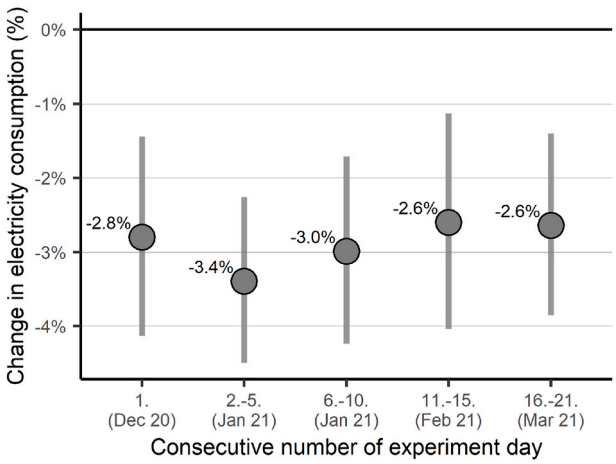


Fig. 8. Average hourly price response in peak hours with a 95% confidence interval in dependency on the consecutive number of the experiment days (the month when most of these days occurred in parentheses). Model results can be found in appendix Table 13.

Table 3. Without checking these web pages, the experiment participants could not know which hours had peak prices, but they would still know that the next day was an experiment day since they were informed via SMS or push message. These results seem to support response fatigue, counterbalanced by learning effects and long-lasting behaviour changes that reduce electricity consumption regardless of an experiment day. As a result, the response remained stable over time. This finding is consistent with other research studies that have reported relatively consistent responses over time (Bartusch and Alvehag, 2014; CER, 2011; Gillan, 2017; Ito et al., 2018; Parrish et al., 2019; Stromback et al., 2011) but contradicts the results of a dynamic pricing pilot which saw strong response fatigue for manual response (Vanthournout et al., 2015).

Responding to the price signals can lead to a long-lasting change in electricity consumption behaviour even on days without price signals. As shown in Table 6, the electricity consumption is reduced by 1.57% on non-experiment days compared to 2.92% in peak hours on experiment days. This reduction on non-experiment days indicates a change in the households' consumption habits and that measures, such as reduced indoor temperature, were maintained over prolonged periods. Similar effects on the consumption habit on days without price signals have been reported by other studies which tested dynamic pricing (Garnache et al., 2021; Martin and Rivers, 2018). Consequently, more than half of the reduction in peak hours may be due to long-lasting changes in consumption behaviour triggered by the price signals and information provided to households on how to react to them. On the other hand, less than half of the response in peak hours may be related to an immediate response to the price signals on a specific experiment day.

4.6. Hourly response stability in peak hours

Another question is if the hourly response is stable throughout the peak period, especially for longer durations, such as in profile B with a peak price period of 11 h. Results from model (2) with an interaction term of the profile and the hour of the day on experiment days D_{it}^{exp}, show a varying hourly response between approximately -2 to -3.5%, as visualised in Fig. 9 a). The most considerable reductions occur from 9 a.m. to 1 p.m. and the lowest from 4 p.m. to 6 p.m. These results indicate that households' flexibility is different over the day. Daily routines, such as people returning home from work after 4 p.m., seem to reduce flexibility and lead to a lower response, whereas it is easier to reduce consumption in the hours after 9 a.m. when people are usually at work and the discomfort of measures such as a reduced indoor temperature is lower. These results are in contrast to previous research which found a

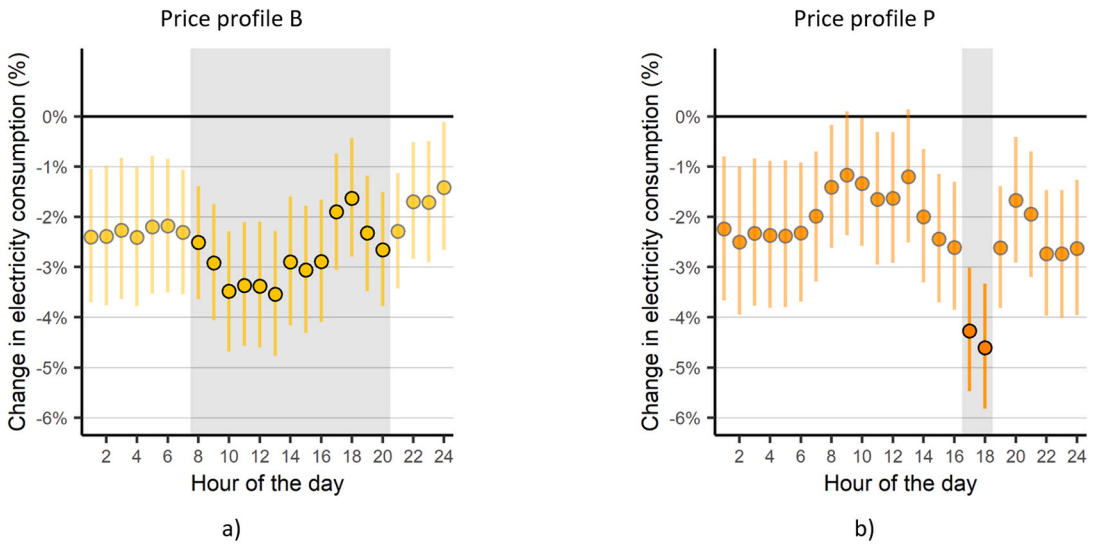


Fig. 9. Average hourly price response with a 95% confidence interval in each hour of the day on experiment days for a) price profile B, and b) profile P. Peak hours are marked in grey and model results can be found in appendix Table 15.

higher response in hours when people usually are at home such as the late afternoon and early evening hours and can manually adjust their electricity consumption (Zhou et al., 2017). A definite answer of the underlying cause for the variations in the hourly response cannot be provided since we have no data which could clarify this question.

4.7. Rebound effect

Shifting of electricity consumption, which has been a popular measure according to the survey answers in Fig. 3, may result in a rebound effect with higher electricity consumption right before or after peak hours. As presented in Fig. 9 for profiles B and P, the hourly response results show a reduced consumption for all hours on experiment days, including off-peak hours. Thus, a rebound effect cannot be observed in our results. This might be again explained by long-lasting measures such as reducing the indoor temperature, which may be in effect for the whole day. In addition, the load shifting was performed mainly with dishwasher, washing machine or tumble dryer, which all have a relative low electricity demand compared to electrical heating. Some hours of the day show less response; for example, the morning hours and 7 p.m.–9 p.m. in profile P and could therefore indicate a partial load shift to these hours. Generally, a rebound is unlikely for manually performed measures since it would require turning on appliances synchronised across households after peak hours, and households may not be willing to put such an effort into such an exact timing. The result of no rebound and reduced electricity consumption outside the peak hours is supported by previous research (Garnache et al., 2021; Ito et al., 2018; VaasaETT, 2017). However, it is in contrast to the results of other studies who found a rebound effect due to load shifting (Azarova et al., 2020; CER, 2011; Schofield et al., 2015).

5. Validity of the results and further work

The experimental design of our study has affected the price response results in both directions. On the one hand, the observed demand reduction results may underestimate the real potential due to the effect of loss aversion (Hobman et al., 2016; Kahneman et al., 1991; Wilkinson and Klaes, 2017), where losses are valued higher than earnings, and experiment participants were rewarded for reducing their electricity consumption. In contrast, a response to variable power prices or grid tariffs would be motivated by avoiding a high bill and should give a larger incentive. This explanation is supported by the evidence that pricing programmes based on rebates perform worse than similar programmes where the price has to be paid (Stromback et al., 2011).

On the other hand, two characteristics of our pricing experiment should lead to higher demand reduction results: self-selection bias and highly electrified participants. Our pricing experiment was conducted with voluntary participants, leading to a self-selection bias towards households interested in electricity consumption and thus possible higher demand response. However, the comparison of household characteristics of experiment participants with a representative sample of the general population, see Table 2, showed only minor differences. Furthermore, as presented in the same table, only 7% of the population answered that they cannot or do not want to reduce their electricity consumption in hours with high electricity prices, compared to 3% in the experiment households. Therefore, we might assume that the observed reduction results in the experiment are only marginally overestimated. However, Norwegian households have highly electrified energy use, which increases demand flexibility and should lead to a higher response. Results from other research support this hypothesis by reporting higher responses from households with electric heating (Parrish et al., 2019; Reiss and White, 2005; Yan et al., 2018) or electric vehicles (Garnache et al., 2021; Gattaciecca et al., 2020).

More research on the flexibility of vulnerable households and an increased understanding of low-flexibility households is needed, and the available information in our dataset can support this work (Hofmann and Siebenbrunner, 2023). The response of the households in this pricing experiment was manually performed. Further automation of demand response, for example, smart charging of electric vehicles, will inevitably lead to a higher price response (Gattaciecca et al., 2020; Parrish et al., 2019) but may also lead to rebound effects and even the creation of new demand peaks due to load synchronisation (Lee and Max Zhang, 2022). Further work should investigate the role of automatic response and its potential to increase demand response to price signals while avoiding rebound effects. Moreover, the pricing experiment was finished in the spring of 2021. From August 2021, electricity prices have skyrocketed in Europe and Norway, reaching levels comparable to those applied in the experiment price signals. Comparing the experimental results with the price response in this situation could bring new knowledge about the generalisability of experiments to real-world settings.

Table 7
Summary of the answers to the research questions.

Research question	Empirical result
Q1 a)	• Response with load reduction, mainly related to space heating, and load shifting of activities
Q1 b)	• Significant average response of 0.085 kWh/h or 2.92% in peak hours
Q2 a)	• Prices over 15 NOK/kWh increase the response, below this value no significant increase • Short peak duration of 2 h increases the response • Slight indication, but not significant, that a lower frequency of events increases response
Q2 b)	• No response above 5 °C, a slight indication of a higher percentage response between -10 and 0 °C
Q2 c)	• Persistent response over time with a slight indication of higher response on experiment days two to five • Long-lasting reduction in electricity consumption on non-experiment days of 1.57%
Q2 a)	• Varying hourly response in peak hours dependent on the hour of the day
Q2 b)	• No rebound effect

6. Conclusions and policy implications

A pricing experiment with 3746 Norwegian households and 14 different price signals was conducted to reveal the potential of implicit demand flexibility from highly electrified end-users in the winter season of 2020/21. This experiment was implemented with a peak rebate program which gave households rewards according to their response and the prices on experiment days. A thorough analysis of the collected hourly electricity consumption data led to the following main findings to the formulated research questions as summarised in Table 7.

The results support earlier findings that households can respond manually to price signals. Furthermore, we find the aggregated manual response from households to be persistent over time due to only minor response fatigue. Consequently, we recommend power system operators and other stakeholders to trust and plan with implicit demand flexibility from households even without automatic demand response in place. In conclusion, the price responsiveness of households should be considered in demand prognoses and energy system planning. An interesting side effect of our pricing experiment was reduced overall electricity consumption on non-experiment days. Pricing experiments or demand-response programmes could therefore also incentivise and educate for long-term changes in energy use behaviour.

The magnitude of the observed manual price response may have limited practical relevance for alleviating grid congestions or for large-scale balancing of renewable energy production. The response was only a few percentage points and in addition only around half of it can be attributed to hourly changes in electricity price. Hence, households' price response and their demand flexibility must be increased in order to be able to play their envisaged active role in a future climate-neutral energy system.

To achieve this, we have the following policy recommendations. First of all, households should be exposed to price signals by implementing and promoting dynamic electricity price contracts, grid tariffs or demand response programs with rewards on the electricity bill that reflect the challenges of the power system. If these price signals are established, policies should focus on enhancing price information to the households and providing information about the economic advantages of reacting to variable prices, enabling informed decisions on whether to manually respond to these prices or invest in automated demand response. Individualised price information should be readily accessible and include the total price of electricity, encompassing all price signals. This requires a standardised format for sharing information about electricity price contracts and grid tariffs. Regulators should be tasked with defining this format and implementing regulations that ensure the availability of such information to third parties with the household's consent. When coupled with available hourly demand data, it would also enable the calculation of potential savings through manual response or the profitability of investments in smart appliances or energy storage on an individualised basis.

Furthermore, increased automatic response or energy storage will help to utilise the full potential of implicit demand flexibility. Price signals should serve as an incentive for investing in such devices but if the profitability of the investments is too low or uncertain for households, policy options such as investment support schemes could be used to increase the uptake.

CRediT authorship contribution statement

Matthias Hofmann: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Karen Byskov Lindberg:** Conceptualization, Methodology, Formal analysis, Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data related to this article are published at Zenodo and described in a data article which has been accepted of Data in Brief.

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Appendix

Table 8
Comparison of electricity consumption between control and treatment groups in each region.

Region	Mean electricity consumption in kWh/h (number of households)			
	Control group	Treatment group	Difference	p-value
Bergen	2.52 (111)	2.25 (218)	0.27	0.085
Bodø	2.24 (214)	2.26 (221)	-0.02	0.896
Oslo	2.34 (257)	2.22 (1530)	0.12	0.271
Tromsø	2.48 (266)	2.53 (524)	-0.05	0.665
Trondheim	1.98 (204)	2.04 (201)	-0.06	0.690

***p < 0.001, **p < 0.01, *p < 0.05.
Note: The p-value of the difference was calculated with Welch's *t*-test.

Table 9
Absolute and percentage changes in electricity consumption in peak hours for each price signal.

Variable	Change (kWh/h)		Change (%)		Mean consumption E_β (kWh/h)
	Coefficient estimate β	95% confidence interval	Estimate	95% confidence interval	
Non-treatment hours	-0.039**	[-0.063, -0.014]	-1.57**	[-2.52, -0.57]	2.440
Non-peak hours	-0.058***	[-0.089, -0.027]	-2.16***	[-3.27, -1.02]	2.631
A_5	-0.082***	[-0.123, -0.04]	-2.91***	[-4.3, -1.44]	2.737
A_10	-0.086**	[-0.14, -0.033]	-2.66**	[-4.26, -1.04]	3.148
B_2	-0.068***	[-0.1, -0.036]	-2.43***	[-3.53, -1.3]	2.73
B_5	-0.081***	[-0.119, -0.043]	-2.76***	[-4.01, -1.49]	2.851
B_10	-0.083***	[-0.119, -0.047]	-2.93***	[-4.14, -1.68]	2.753
B_15	-0.095***	[-0.14, -0.051]	-3.19***	[-4.63, -1.74]	2.882
C_5	-0.083***	[-0.124, -0.042]	-3.14***	[-4.62, -1.62]	2.558
C_10	-0.092***	[-0.139, -0.045]	-3.39***	[-5.03, -1.69]	2.625
P_2	-0.096***	[-0.143, -0.05]	-3.16***	[-4.63, -1.67]	2.946
P_5	-0.154***	[-0.213, -0.095]	-4.56***	[-6.2, -2.86]	3.225
P_10	-0.139***	[-0.183, -0.095]	-4.38***	[-5.69, -3.04]	3.033
P_15	-0.188***	[-0.243, -0.132]	-5.71***	[-7.26, -4.08]	3.106
P0_2	-0.122***	[-0.187, -0.056]	-4.21***	[-6.32, -1.98]	2.773
P0_10	-0.117***	[-0.173, -0.062]	-3.82***	[-5.54, -2.06]	2.949
P0_30	-0.199***	[-0.272, -0.126]	-6.69***	[-8.93, -4.34]	2.775

***p < 0.001, **p < 0.01, *p < 0.05.
Note: Estimates of the various price signals $D_{it}^{psignal}$ are calculated with model (2).

Table 10
Absolute difference of the electricity consumption in peak hours against the respective reference level for various model specifications.

Variable	Coefficient estimate β (kWh/h)					Mean consumption(kWh/h)
Non-treatment hours	-0.039**	-0.039**	-0.039**	-0.039**	-0.039**	2.440
Non-peak hours	-0.058***	-0.058***	-0.058***	-0.058***	-0.058***	2.631
Peak hours	-0.07***	-0.081***	0.016	-0.067***	0.026	2.821
Price 5	-0.019			-0.021*	-0.017	2.862
Price 10	-0.017			-0.017	-0.014	2.814
Price 15	-0.106***			-0.069**	-0.066**	3.106
Price 30	-0.115***			-0.092**	-0.09**	2.775
Profile A		-0.004		0.002	0.002	2.954
Profile C		-0.009		-0.004	-0.006	2.592
Profile P		-0.067***		-0.051**	-0.049**	3.066
Profile P0		-0.058*		-0.04	-0.039	2.861
Temperature (-20,-15]			-0.105		-0.098	3.181
Temperature (-15,-10]			-0.102		-0.094	3.094
Temperature (-10,-5]			-0.111**		-0.105**	3.089
Temperature (-5,0]			-0.108**		-0.104**	2.842
Temperature (0,5]			-0.077*		-0.074*	2.321
Price	X			X	X	
Profile		X		X	X	
Temperature			X		X	

***p < 0.001, **p < 0.01, *p < 0.05.
Note: The estimates are calculated for different versions of model (3), including various combinations of confounding variables of peak price level D_{it}^{price}, price profile $D_{it}^{profile}$ and outdoor temperature D_{it}^{temp}. The reference levels are for the price level: 2 NOK/kWh (Price 2), for the price profile: Profile B, and for the outdoor temperature: temperature range of 5 to 10 °C (Temperature (5,10]).

Table 11
Percentage difference of the electricity consumption in peak hours against the respective reference level for various model specifications.

Variable	Coefficient estimate divided by the corrected average demand in peak hours (%)				
Price 5	-0.65			-0.72*	-0.58
Price 10	-0.58			-0.58	-0.48
Price 15	-3.65***			-2.37**	-2.27**
Price 30	-3.96***			-3.17**	-3.10**
Profile A		-0.14		0.07	0.07
Profile C		-0.31		-0.14	-0.21
Profile P		-2.31***		-1.75*	-1.69**
Profile P0		-2.00*		-1.38	-1.34
Temperature (-20,-15]			-3.61		-3.37
Temperature (-15,-10]			-3.51		-3.23
Temperature (-10,-5]			-3.82**		-3.61**
Temperature (-5,0]			-3.72**		-3.58**
Temperature (0,5]			-2.65*		-2.55*
Price	X			X	X
Profile		X		X	X
Temperature			X		X

***p < 0.001, **p < 0.01, *p < 0.05.

Note: The percentage values are calculated based on the results in Table 10. The absolute values are divided by the average demand in peak hours corrected for the estimated reduction (for example: 2.906 = 2.821 + 0.085) as described in Chapter 3.

Table 12
Absolute and percentage changes in electricity consumption in peak hours for different outdoor temperature ranges.

Variable	Change (kWh/h)		Change (%)		Mean consumption E_β (kWh/h)
	Coefficient estimate β	95% confidence interval	Estimate	95% confidence interval	
Non-treatment hours	-0.039**	[-0.063, -0.014]	-1.57**	[-2.52, -0.57]	2.440
Non-peak hours	-0.058***	[-0.089, -0.027]	-2.16***	[-3.27, -1.02]	2.631
H_peak_(-20,-15]	-0.089	[-0.261, 0.083]	-2.72	[-7.58, 2.68]	3.181
H_peak_(-15,-10]	-0.086	[-0.175, 0.003]	-2.7	[-5.35, 0.1]	3.094
H_peak_(-10,-5]	-0.096***	[-0.137, -0.054]	-3.01***	[-4.25, -1.72]	3.089
H_peak_(-5,0]	-0.093***	[-0.126, -0.059]	-3.17***	[-4.25, -2.03]	2.842
H_peak_(0,5]	-0.061***	[-0.09, -0.032]	-2.56***	[-3.73, -1.36]	2.321
H_peak_(5,10]	0.016	[-0.044, 0.076]	0.77	[-2.07, 3.78]	2.086

***p < 0.001, **p < 0.01, *p < 0.05.

Note: Estimates of the effect of outdoor temperature D_{it}^{temp} are calculated with model (2).

Table 13
Absolute and percentage changes in electricity consumption in peak hours for different consecutive numbers of the experiment day.

Variable	Change (kWh/h)		Change (%)		Mean consumption E_β (kWh/h)
	Coefficient estimate β	95% confidence interval	Estimate	95% confidence interval	
Non-treatment hours	-0.039**	[-0.063, -0.014]	-1.57**	[-2.52, -0.57]	2.440
Non-peak hours	-0.058***	[-0.089, -0.027]	-2.16***	[-3.27, -1.02]	2.631
Expday 1	-0.071***	[-0.106, -0.036]	-2.8***	[-4.13, -1.44]	2.461
Expday 2-5	-0.1***	[-0.134, -0.066]	-3.39***	[-4.49, -2.26]	2.853
Expday 6-10	-0.094***	[-0.135, -0.053]	-2.99***	[-4.24, -1.71]	3.05
Expday 11-15	-0.082***	[-0.129, -0.035]	-2.6***	[-4.04, -1.13]	3.068
Expday 16-21	-0.065***	[-0.096, -0.034]	-2.64***	[-3.85, -1.4]	2.395

***p < 0.001, **p < 0.01, *p < 0.05.

Note: Estimates of the effect of the consecutive number of experiment day D_{it}^{expday} are calculated with model (2).

Table 14
Absolute and percentage changes in electricity consumption in peak hours for different duration since the last experiment day.

Variable	Change (kWh/h)		Change (%)		Mean consumption E_β (kWh/h)
	Coefficient estimate β	95% confidence interval	Estimate	95% confidence interval	
Non-treatment hours	-0.039**	[-0.063, -0.014]	-1.57**	[-2.52, -0.57]	2.440
Non-peak hours	-0.058***	[-0.089, -0.027]	-2.16***	[-3.27, -1.02]	2.631
Days First	-0.071***	[-0.106, -0.036]	-2.8***	[-4.13, -1.44]	2.461
Days 0	-0.08***	[-0.117, -0.042]	-2.66***	[-3.84, -1.42]	2.926
Days 1-6	-0.087***	[-0.122, -0.053]	-2.94***	[-4.07, -1.81]	2.876
Days 7-19	-0.092***	[-0.124, -0.06]	-3.45***	[-4.59, -2.27]	2.578

***p < 0.001, **p < 0.01, *p < 0.05.

Note: Estimates of the effect of the duration since the last experiment day D_{it}^{days} are calculated with model (2).

Table 15
Absolute and percentage changes in electricity consumption in each hour of the day for profiles B and P.

Variable	Change (kWh/h)		Change (%)		Mean consumption E_β (kWh/h)
	Coefficient estimate β	95% confidence interval	Estimate	95% confidence interval	
Non-treatment hours	-0.039**	[-0.063, -0.014]	-1.57**	[-2.52, -0.57]	2.440
Non-peak hours	-0.058***	[-0.089, -0.027]	-2.16***	[-3.27, -1.02]	2.631
B 01	-0.06***	[-0.094, -0.026]	-2.4***	[-3.7, -1.05]	2.444
B 02	-0.057***	[-0.091, -0.023]	-2.39***	[-3.76, -0.98]	2.329
B 03	-0.053**	[-0.086, -0.019]	-2.27**	[-3.64, -0.83]	2.278
B 04	-0.056***	[-0.089, -0.023]	-2.41***	[-3.78, -1.01]	2.263
B 05	-0.051**	[-0.083, -0.018]	-2.2**	[-3.53, -0.79]	2.27
B 06	-0.052**	[-0.085, -0.02]	-2.18**	[-3.51, -0.85]	2.336
B 07	-0.06***	[-0.093, -0.027]	-2.31***	[-3.54, -1.06]	2.532
B 08	-0.073***	[-0.107, -0.04]	-2.51***	[-3.64, -1.39]	2.832
B 09	-0.086***	[-0.121, -0.051]	-2.92***	[-4.06, -1.75]	2.862
B 10	-0.1***	[-0.136, -0.065]	-3.48***	[-4.68, -2.29]	2.771
B 11	-0.094***	[-0.129, -0.058]	-3.37***	[-4.57, -2.11]	2.693
B 12	-0.093***	[-0.128, -0.057]	-3.38***	[-4.6, -2.1]	2.655
B 13	-0.096***	[-0.131, -0.061]	-3.54***	[-4.77, -2.28]	2.615
B 14	-0.077***	[-0.112, -0.042]	-2.9***	[-4.16, -1.6]	2.579
B 15	-0.082***	[-0.117, -0.047]	-3.06***	[-4.31, -1.78]	2.599
B 16	-0.081***	[-0.116, -0.046]	-2.89***	[-4.09, -1.66]	2.723
B 17	-0.057**	[-0.093, -0.022]	-1.9**	[-3.06, -0.74]	2.942
B 18	-0.05**	[-0.087, -0.013]	-1.63**	[-2.79, -0.43]	3.026
B 19	-0.072***	[-0.109, -0.036]	-2.32***	[-3.48, -1.18]	3.026
B 20	-0.082***	[-0.118, -0.046]	-2.66***	[-3.78, -1.51]	3.006
B 21	-0.07***	[-0.106, -0.034]	-2.29***	[-3.43, -1.13]	2.987
B 22	-0.051**	[-0.086, -0.015]	-1.7**	[-2.84, -0.51]	2.944
B 23	-0.049**	[-0.084, -0.014]	-1.71**	[-2.9, -0.49]	2.817
B 24	-0.038*	[-0.072, -0.003]	-1.42*	[-2.66, -0.11]	2.636
P 01	-0.057**	[-0.095, -0.02]	-2.24**	[-3.67, -0.8]	2.493
P 02	-0.061***	[-0.098, -0.024]	-2.5***	[-3.95, -1]	2.383
P 03	-0.056**	[-0.092, -0.02]	-2.33**	[-3.77, -0.84]	2.349
P 04	-0.057**	[-0.093, -0.021]	-2.37**	[-3.81, -0.89]	2.348
P 05	-0.058**	[-0.094, -0.021]	-2.38**	[-3.8, -0.88]	2.379
P 06	-0.059***	[-0.095, -0.023]	-2.32***	[-3.69, -0.92]	2.48
P 07	-0.055**	[-0.092, -0.019]	-1.99**	[-3.29, -0.7]	2.708
P 08	-0.043*	[-0.081, -0.005]	-1.41*	[-2.62, -0.17]	3.011
P 09	-0.036	[-0.074, 0.003]	-1.17	[-2.37, 0.1]	3.042
P 10	-0.04*	[-0.078, -0.001]	-1.34*	[-2.58, -0.03]	2.945
P 11	-0.048*	[-0.087, -0.009]	-1.65*	[-2.95, -0.31]	2.867
P 12	-0.047*	[-0.085, -0.009]	-1.63*	[-2.92, -0.32]	2.83
P 13	-0.034	[-0.072, 0.004]	-1.2	[-2.51, 0.14]	2.798
P 14	-0.056**	[-0.094, -0.018]	-2**	[-3.31, -0.65]	2.747
P 15	-0.069***	[-0.106, -0.032]	-2.44***	[-3.7, -1.15]	2.762
P 16	-0.077***	[-0.115, -0.038]	-2.61***	[-3.85, -1.3]	2.875
P 17	-0.135***	[-0.175, -0.094]	-4.27***	[-5.47, -3.01]	3.027
P 18	-0.15***	[-0.192, -0.107]	-4.61***	[-5.82, -3.33]	3.105
P 19	-0.086***	[-0.127, -0.045]	-2.62***	[-3.82, -1.39]	3.194
P 20	-0.054**	[-0.095, -0.013]	-1.67**	[-2.91, -0.41]	3.172
P 21	-0.062**	[-0.103, -0.022]	-1.95**	[-3.2, -0.7]	3.115
P 22	-0.085***	[-0.125, -0.045]	-2.74***	[-3.97, -1.47]	3.021
P 23	-0.081***	[-0.12, -0.043]	-2.74***	[-4.01, -1.47]	2.875
P 24	-0.072***	[-0.11, -0.034]	-2.63***	[-3.96, -1.26]	2.668

***p < 0.001, **p < 0.01, *p < 0.05.
Note: Estimates for the response in each hour of the day for profiles B and P are calculated with model (2).

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